Expanding Double Brackets

Exam code: 0607

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Written by: Mark Curtis
Reviewed by: Dan Finlay
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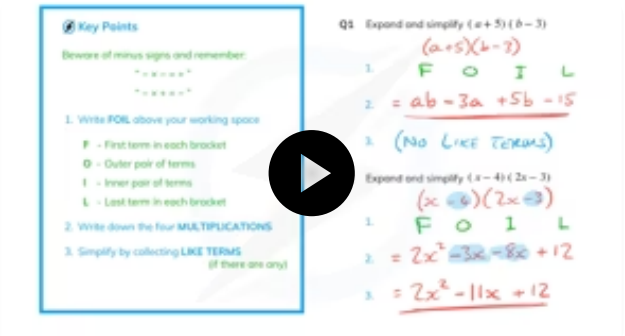

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Expanding two brackets

How do I expand two brackets?

- **Every term** in the **first** bracket must be multiplied by **every term** in the **second** bracket
  - Expanding $(x+1)(x+3)$ requires 4 multiplications in total
    - $x(x+1)+3(x+1)$
    - $x \times x + 1x + 3x + 3$
    - $x^2 + 4x + 3$

Using a grid

- You can use a **grid** for a visual way to expand brackets
- To expand $(x+1)(x+3)$, write out the brackets as **row** and **column headings** of a grid
  - They can be in either direction
  - Remember to write the appropriate **sign** in front of each term

	x	+1
x		
+3		

Help me understand
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

x	x^2	x
+3	$3x$	3

- **Add** together all the terms **inside** the grid to get the answer
  - $x^2 + x + 3x + 3$

- Collect **like** terms
- $x^2 + 4x + 3$

Using FOIL

- **Every term** in the **first** bracket must be multiplied by **every term** in the **second** bracket
  - Expanding $(x + 1)(x + 3)$ requires 4 multiplications in total

- A good way to remember all the multiplications is **FOIL**
  - **F = First**: multiply together the **first** terms in each bracket
  - **O = Outside**: multiply the first term in the first bracket by the last term in the last bracket
    - Visually, these are the **outer** terms

  - **I = Inside**: multiply the last term in the first bracket by the first term in the last bracket
    - Visually, these are the **inner** terms

  - **L = Last**: multiply together the **last** terms in each bracket

- It helps to put **negative** terms in **brackets** when multiplying
- **Simplify** the final answer by **collecting like terms** (if there are any)

How do I expand when there are multiple variables?

- All the same rules and methods apply as when there is just one variable
- Remember to only simplify **like terms**
- For example: $(3x + 2y)(4x - 6y)$
  - Expanding: $12x^2 - 18xy + 8xy - 12y^2$
  - The xy terms can be combined
  - $12x^2 - 10xy - 12y^2$

Worked Example

(a) Expand $(2x - 3)(x + 4)$.

Answer:

$$2x \times x + 2x \times 4 + (-3) \times x + (-3) \times 4$$

Simplify each term

$$2x^2 + 8x - 3x - 12$$

Collect like terms (the 8*x* and -3*x*)

$$2x^2 + 5x - 12$$

(b) Expand $(x-3)(3x-5)$.

Answer:

Using FOIL, multiply together the first, outer, inner and last terms

$$\overset{\text{F}}{x \times 3x} + \overset{\text{O}}{x \times (-5)} + \overset{\text{I}}{(-3) \times 3x} + \overset{\text{L}}{(-3) \times (-5)}$$

Simplify each term

$$3x^2 - 5x - 9x + 15$$

Collect like terms (the -5*x* and -9*x*)

$$3x^2 - 14x + 15$$

Worked Example

Expand $(3r+2t)(5t-8r)$.

Answer:

Expand using your chosen method, here we will use a grid

	$3r$	$+2t$
$5t$		
$-8r$		

Work out the term in each place in the grid by multiplying

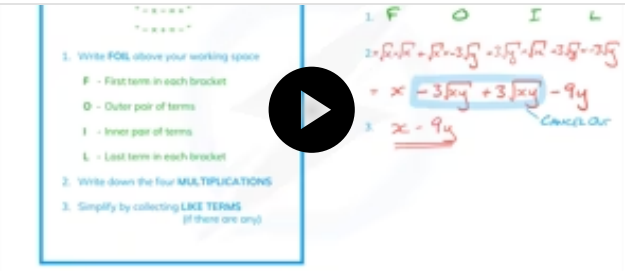
	$3r$	$+2t$
$5t$	$15rt$	$10t^2$
$-8r$	$-24r^2$	$-16rt$

So the expanded expression is

$$10t^2 + 15rt - 16rt - 24r^2$$

The rt terms can be combined

$$10t^2 - rt - 24r^2$$



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Expanding squared brackets

How do I expand a bracket squared?

- Remember that a **square** number is a number multiplied by itself
- Write $(x + 3)^2$ as $(x + 3)(x + 3)$ and use one of the methods above
  - With FOIL: $(x + 3)(x + 3) = x^2 + 3x + 3x + 9$
  - Then collect like terms: $x^2 + 6x + 9$

Examiner Tips and Tricks

Do not make the common mistake of saying $(x + y)^2$ is the same as $x^2 + y^2$. In general, this is not true.

Worked Example

Expand $(2x + 3)^2$.

Answer:

Remember that the answer is not $(2x)^2 + 3^2$
Rewrite the expression as two separate brackets multiplied together

$$(2x + 3)(2x + 3)$$

Using FOIL, multiply together the first, outer, inner and last terms

$$\overset{\text{F}}{2x \times 2x} + \overset{\text{O}}{2x \times 3} + \overset{\text{I}}{3 \times 2x} + \overset{\text{L}}{3 \times 3}$$

Collect like terms (the $6x$ and $6x$)

$$4x^2 + 12x + 9$$

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